

Supplementary document

Imagine for a moment our world without any electricity. That's unthinkable, isn't it? Welcome to this exploration where we tackle a real scientific monument. Today, we will dive into the heart of the electric current, to understand exactly how the invisible manages to power our daily lives. The journey promises to be fascinating, from the observation of the microscopic world of atoms to the construction, step by step, of a real functional stack. So, here's the revelry program. We will start by looking at the mystery of the electric current itself, before following the incredible journey of electrons. Then, instead of practice, we will see how we can make an authentic volta battery with our own hands. This will allow us to unveil its secret, the famous oxidation-reduction, to finally lead to the fascinating technological evolution of modern batteries. Come on, first part, we attack directly with the mystery of the electric current. And an unexpected little question. The question is asked. Can you seriously light a lamp using just a few pieces of metal and a little salt water? Frankly, put like that, it almost sounds like a magic trick, or bad science fiction. But in reality, it is an absolutely fundamental scientific principle. And that's what we're going to dissect together. To understand the trick, let's take a very simple everyday object. A flashlight circuit. We have a generator, the battery, which acts as a kind of motor to push a direct current into a closed loop of conductive wires. We press the switch, the circuit closes, and paf! The lamp filament where it heats up is produced light. Everyone knows that. But the real fundamental question is where does this famous current inside the metal tip come from? And that takes us straight to our second step. We will be interested in the journey of the infinitely small. To understand, you have to radically change your dimension. If we zoom inside the copper wire of our flashlight, but be careful, a very, very powerful microscopic zoom, we discover that it is not just an inert solid object, it is actually an ultra-organized tangle of atoms. And each atom is composed of a positive central core surrounded by a huge cloud of particles that keeps moving. These tiny frenetic particles are called electrons. They carry a negative charge, and in a normal, say neutral atom, they perfectly balance the positive charges of the nucleus. It is simply the absolute basis of all the matter that surrounds us. But then, why do we insist on using copper for our cables? Well, that's because copper has a real superpower. Of all its 29 electrons, only one lies on the outermost layer, called the valence layer. And the rule in chemistry is that if a material has less than 4 electrons on that layer, it's an excellent conductor. As a result, our small solitary copper electron is not very well attached to its core. It becomes what is called a free electron that can move quite easily from one atom to another. And that's where the magic happens. When the lamp is turned on, the generator imposes a voltage. No more chaos! Immediately, these free electrons stop moving in all directions and start jumping from atom to atom in an extremely coordinated movement. And this ultra-ordered movement of free electrons, well that's exactly what electric current is. No more complicated than that. There's a perfect analogy for imagining this in action. Let's take the example of a simple garden hose, which is already full of water. As soon as you turn on the tap, the pressure pushes the water out the other side almost instantly. However, the specific water molecules that have just come out of the tap, they will take quite a long time to cross the entire length of the pipe. And this analogy puts the finger on something quite bluffing. A very common confusion. The electromagnetic wave, the signal itself, travels at a completely crazy speed, close to that of light. That's why the bulb lights up at a quarter turn. But the little physical electrons, they drag themselves. They advance at the speed of a snail, barely a few centimeters per hour. So, the electrons that heat the filament of the bulb at the moment it is turned on, are not at all those that leave the switch. They were already there in the thread. Well, now that we have understood the journey of the electron, let's move on to the third part. How do we create this current ourselves by making a Volta battery? A small leap back in time. It's the year 1800, and Alessandro Volta has just dropped a real scientific bomb. At the time, after Luigi Galvani's experiments on frog legs, everyone thought that electricity came from biological tissues. We state the balls and prove the opposite. We don't need animals. We can generate a direct current, purely and simply with chemistry and vulgar metals. The crazy thing is that this historical invention can be reproduced super easily on a corner of the table. Simply place a zinc washer. On top, add a piece of blotting paper that has been thoroughly soaked in salt water. Next, we place a copper washer. This exact stacking is repeated six times in a row. Which explains the origin of the word pile. And bingo! This assembly generates enough voltage to turn on a small led bulb. But, let's take a little break. Why specifically salt water? Wouldn't pure water do the trick? Well no! Because pure water is actually an electrical insulator. These molecules are strongly linked together. There are no free electrons on the horizon. On the other hand, by adding salt, we dissolve sodium chloride, which balances ions in our liquid. And in the water, it is these ions that will do the job of carrying the electric current and act as an indispensable bridge between our metal washers. And that brings us straight to our fourth part. To reveal the real hidden motor of the battery. Oxido-reduction.

This word is a little scary, but the mechanics below are super logical. To summarize, we have real teamwork. On the one hand, zinc acts as a negative terminal, the anode. It will undergo oxidation, that is, it sacrifices and loses its free electrons. On the other side, copper, our positive terminal or cathode, will undergo what is called a reduction. He will happily recover these same electrons. And it is this vital and continual need to exchange particles between the two that creates an incredible pushing force through our outer circuit. Researchers at the Sorbonne have an excellent mental image to imagine all this. Imagine our two metals as simple sponges, connected by a third sponge in the middle, the electrolyte, our famous salt water. Initially, a sponge is flat to crack ion. The other is completely empty. During discharge, ions sneak in and are squeezed from sponge to sponge through the liquid. This great back and forth movement is actually a real chemical breath. And it is this internal breathing that propels our free electrons outside, into the electrical wire. And here we are on our fifth and final step, the incredible evolution and challenges of our modern batteries. Obviously, we didn't stop at Volta's little find in 1800. There was the lead-acid battery in 1859, then the gigantic lithium-ion revolution in the 90s, the one that is now in almost all our devices. Eventually, the goal remained exactly the same for more than two centuries. To manage to store more and more energy, in an ever smaller volume, and for as long as possible. Except that today, we are facing a real technological wall, especially with electric cars. Lithium-ion batteries use a liquid electrolyte, our famous central sponge. And the problem with liquids is that they can grow unwanted microscopic metal filaments, which are called dendrites. If these dendrites grow too big, go through the liquid and touch the other edge, well, that's the guaranteed short circuit. That's why the new star of research is the so-called all-solid-state battery. The idea is to replace the flammable liquid with a completely solid and highly elastic material. No more cracks, we block the dendrite thrust, safety is increased tenfold, and as a bonus, it promises ultra-fast search times. When you take a step back, the path traveled since 1800 forces admiration. We went from simple metal washers, bathing in salt water, and we have a total metrisis of the electron via state-of-the-art nanometric elastic structures. It's completely bluffing. The question that remains unanswered now is how far this perfect domestication of the infinitely small will push the boundaries of engineering. How will this all-solid chemistry end up fundamentally redefining the way we move around tomorrow? This is the next chapter of this great scientific adventure and we are really looking forward to discovering it.